



HALDIA INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTION UNDER MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL)

Paper Code: PCC-CS 302

Paper Name: Formal Language and Automata Theory

Time Allotted: 3 Hours

Full Marks: 70

The figures in the margin indicate full marks

Candidates are required to give their answers in their own words as far as practicable

Group – A

(Multiple Choice Type Questions)

Choose the correct alternatives from the followings:

15 x 1 = 15

1. (i) $\{a^n b^n \mid n \geq 1\}$ is accepted by a PDA
 - a) by null store and also by final state
 - b) by null store but not by final state
 - c) by final state but not by null store
 - d) by none of these
- (ii) $\{a^n b^{2n} \mid n \geq 1\}$ is accepted by
 - a) a finite automata
 - b) a nondeterministic finite automaton
 - c) a pda
 - d) none of these
- (iii) The intersection of a context-free language and a regular language is
 - a) context-free
 - b) regular but not context free
 - c) neither context-free nor regular
 - d) both regular and context-free
- (iv) If a grammar G has three productions $S \rightarrow aSa \mid bSb \mid c$, then
 - a) $abcba$ and $bacab \in L(G)$
 - b) $bcba$ and $abcab \in L(G)$
 - c) $accca$ and $bcccb \in L(G)$
 - d) $acccb$ and $bccca \in L(G)$
- (v) The regular grammar generating $\{a^n : n \geq 1\}$ is
 - a) $(\{S\}, \{a\}, \{S \rightarrow aS\}, S)$
 - b) $(\{S\}, \{a\}, \{S \rightarrow SS, S \rightarrow a\})$
 - c) $(\{S\}, \{a\}, \{S \rightarrow aS\}, S)$
 - d) $(\{S\}, \{a\}, \{S \rightarrow aS, S \rightarrow a\}, S)$
- (vi) $\{a^n b^n c^n \mid n \geq 1\}$ is
 - a) regular
 - b) context-free but not regular
 - c) context-sensitive but not context-free
 - d) none of these.
- (vii) $\{a^n b^n \mid n \geq 1\}$ is
 - a) regular
 - b) context-free but not regular
 - c) a context-sensitive but not context-free
 - d) none of these.
- (viii) The set of all strings over $\{a, b\}$ of even length is represented by the regular expression
 - a) $(ab+aa+bb+ba)^*$
 - b) $(a+b)^*(a^*+b)^*$
 - c) $(aa+bb)^*$
 - d) $(ab+ba)^*$
- (ix) The set of all strings over $\{a, b\}$ of length 4, starting with an a is represented by the regular expression
 - a) $a(a+b)^*$
 - b) $a(ab)^*$
 - c) $(ab+ba)(aa+bb)$
 - d) $a(a+b)(a+b)(a+b)$
- (x) $(0^*1^*)^*$ is the same as
 - a) $(0+1)^*$
 - b) $(01)^*$
 - c) $(10)^*$
 - d) none of these
- (xi) if L is the set of all strings over $\{a, b\}$ containing at least one a , then it is not represented by the regular expression
 - a) $b^*a(a+b)^*$
 - b) $(a+b)^*a(b+a)^*$
 - c) $(a+b)^*ab^*$
 - d) $(a+b)^*a$

(xii) $\{a^{2n} \mid n \geq 1\}$ is represented by the regular expression

- a) $(aa)^*$ b) a^* c) aa^*a d) a^*a^*

(xiii) The set of strings over $\{a,b\}$ having exactly 3b's is represented by the regular expression

- a) a^*bbb b) $a^*ba^*ba^*b$ c) ba^*ba^*b d) $a^*ba^*ba^*ba^*$

(xiv) The set of all strings over $\{a,b\}$ having abab as a substring is represented by

- a) a^*ababb^* b) $(a+b)^*abab(a+b)^*$ c) $a^*b^*ababa^*b^*$ d) $(a+b)^*abab$

(xv) $(a+a^*)^*$ is equivalent to

- a) $a(a^*)^*$ b) a^* c) aa^* d) none

Group – B

(Short Answer Type Questions)

Attempt any three from the followings:

3 x 5 = 15

2. What is Kleene closure and Positive closure? Explain with example.

3. Find Regular expression over $\Sigma = \{a,b\}$ for the languages defined as follows-

i) $L1 = \{a^m b^n \mid m > 0, n > 0\}$

ii) $L2 = \{a^{2n} b^{2m+1} \mid m \geq 0, n \geq 0\}$

4. Using Pumping lemma prove that $L = \{a^n b^n c^n \mid n \geq 0\}$ is not Context Free.

5. prove $t+RR^*=R^*$ where t is null string and R is a regular expression

6. Using Pumping lemma prove that $L = \{a^{i^2} \mid i \geq 1\}$ is not regular.

Group – C

(Long Answer Type Questions)

Attempt any four from the followings:

4 x 10 = 40

7. a) What is push down automata? Write two advantage of push down automata.

b) Design a PDA for accepting a language $\{a^n b^{2n} \mid n \geq 1\}$.

2+2+6

8. a) Explain Greibach Normal Form (GNF) with example.

b) Convert the following grammar into GNF.

$$S \rightarrow AB/BC$$

$$A \rightarrow aB/bA/a$$

$$B \rightarrow bB/cC/b$$

$$C \rightarrow c$$

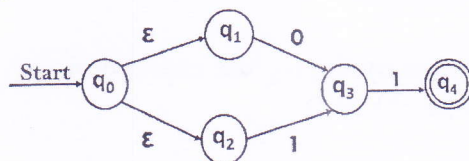
(3+7)

9. a) State the difference between NFA and DFA.

b) Design FA for the language $L = \{(01)^i 1^{2j} \mid i \geq 1, j \geq 1\}$

c) Convert the NFA with ϵ into its equivalent DFA.

2+3+5



10. Let G be the grammar $S \rightarrow SS, S \rightarrow aSb, S \rightarrow bSa, S \rightarrow \lambda$. For the string abab, find leftmost derivation and rightmost derivation. Draw the parse trees for leftmost and rightmost derivations.

5+5

11. What is Turing machine (TM)? Design TM which accepts the language $L = \{aba\}$ over $\Sigma = \{a,b\}$.

2+8

12. What is halting problem? Prove that the halting problem of Turing machine is undecidable.

2+8